

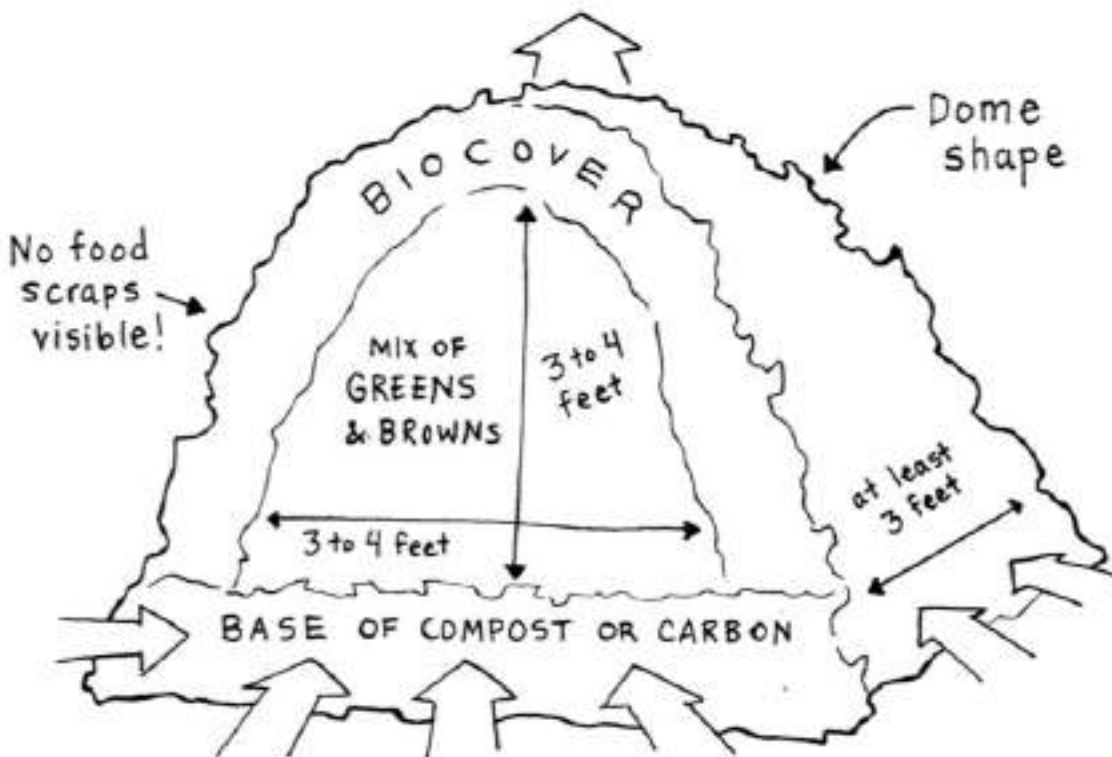
Community Composting 101



Managing the Composting Process

Neighborhood
Soil Rebuilders
COMPOSTER TRAINING PROGRAM

ILSR
INSTITUTE FOR
Local Self-Reliance



Cross-section of properly built composting pile

Managing the Composting Process

✓ During Active Composting:

- Monitoring, Turning, Keeping Records, Troubleshooting

✓ Finishing the Process:

- Curing, Screening, Testing, Using Compost



Ideal Composting Conditions

Variable	Ideal Range
C:N Ratio	25:1 - 30:1
Moisture Content	50 - 60%
Particle Size	1/8 - 2 inches
Bulk Density	800 - 1,000 pounds per cubic yard
Pile Size	> 1 cubic yard

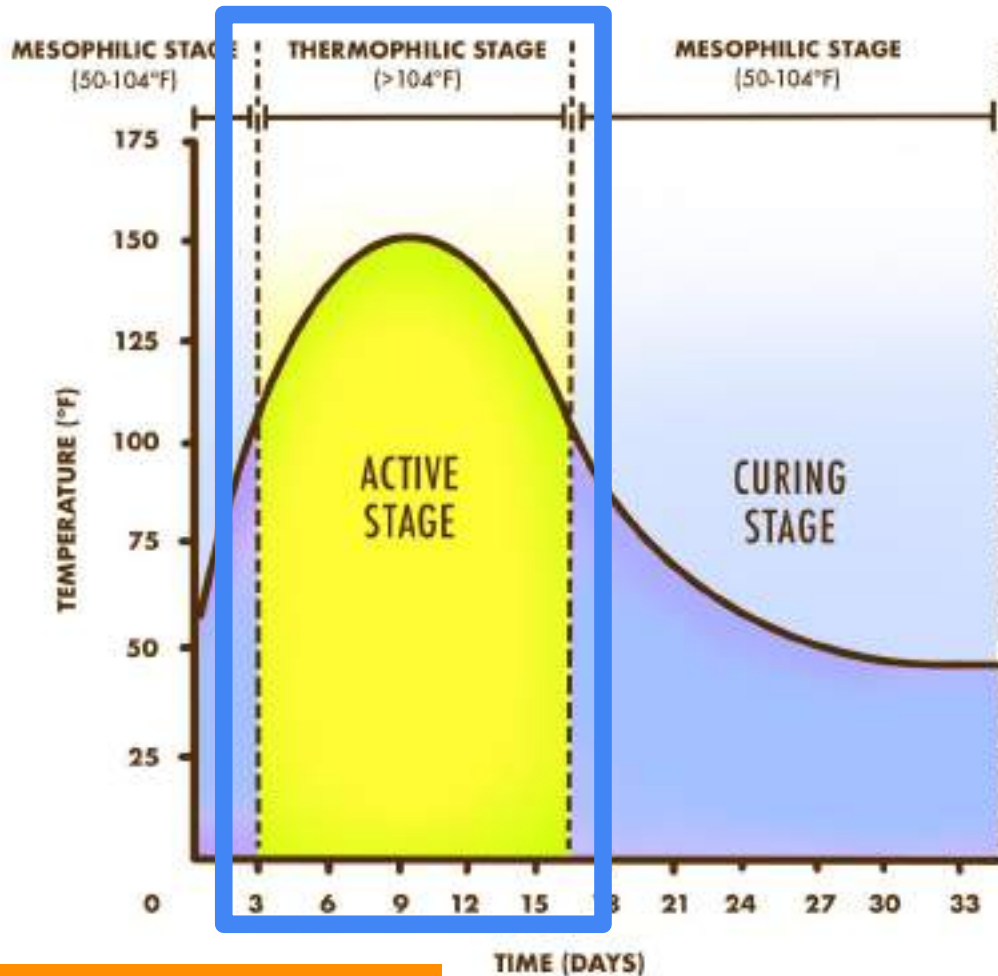
Adapted from the US Composting Council's Compost Facility Operator Training (2015)





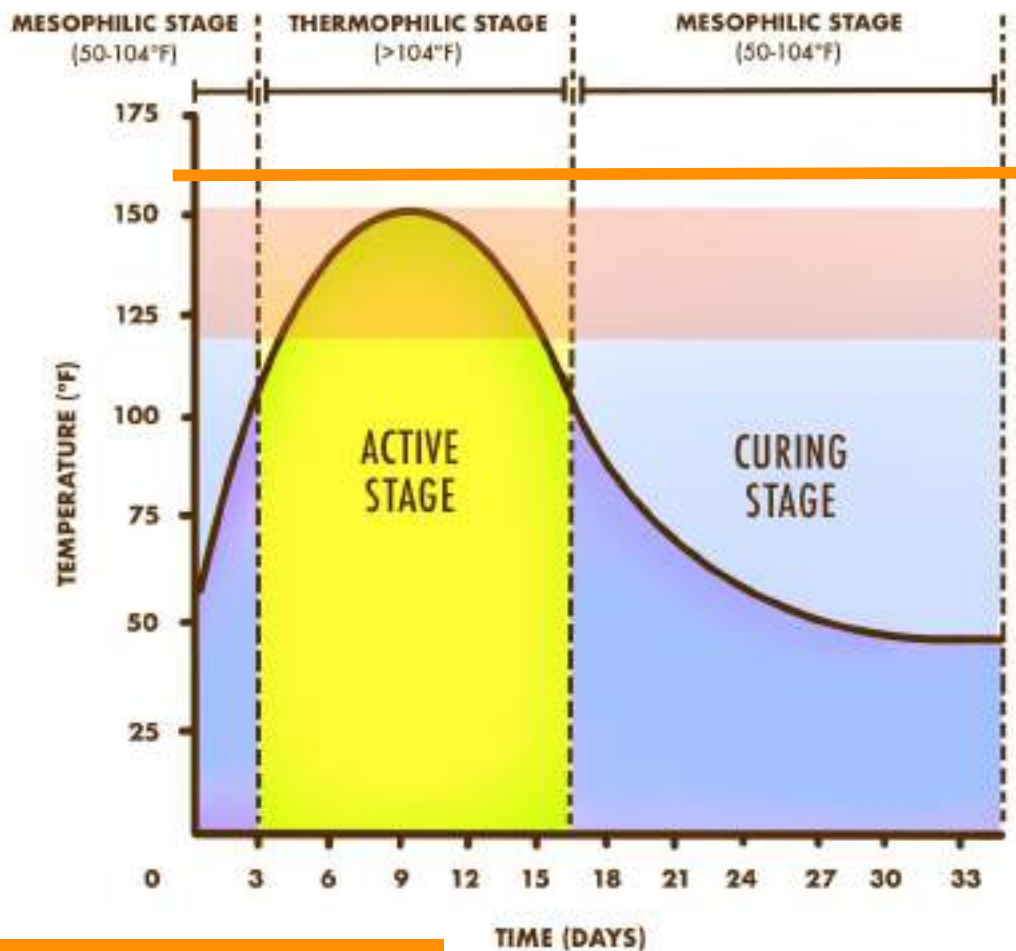
Monitoring

NSR Master Composter Training participants monitor compost pile oxygen levels, taught by Benny Erez of ECO City Farms.



Different Phases, Different Microbes





Different Phases, Different Microbes

Goal	Target Temperature
Achieve Ideal Range	120 - 150 °F
Most Rapid Composting	122 - 140 °F
Pathogen Reduction	≥ 131 °F
Weed Seed Destruction	≥ 145 °F
Tomato Seed Destruction	> 150 °F

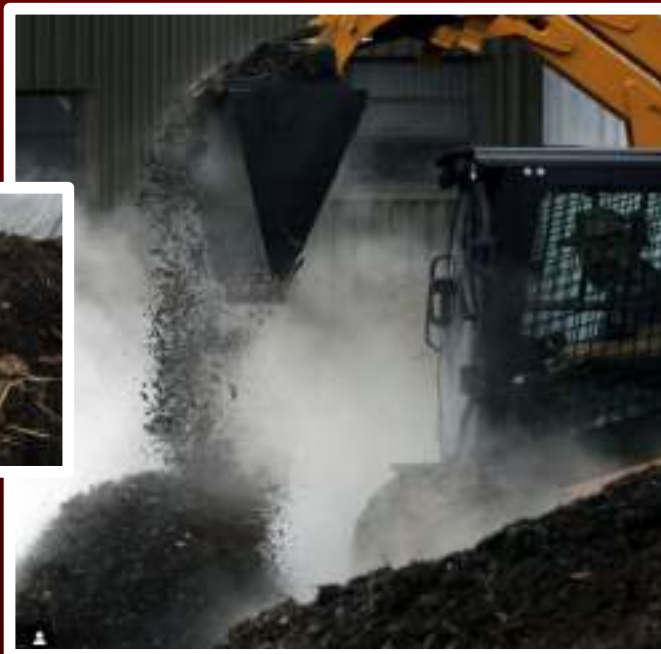
The Process to Further Reduce Pathogens (PFRP)



Produce recalls due to E.coli or Salmonella contamination are common.
Source: USA Today



The Process to Further Reduce Pathogens (PFRP)



Source: @RustBeltRiders @Tilth.Soil



The Process for Reducing Pathogens (PFRP)

Composting System	Measured Temp.	Duration	Notes:
Passively Aerated & Windrows:	$\geq 131^{\circ}\text{F}$ (55°C)	≥ 15 days	During this period: ≥ 5 turnings, with ≥ 3 days in between turnings. Days do not necessarily need to be continuous (see local regulations).
Aerated Piles & In-Vessel:	$\geq 131^{\circ}\text{F}$ (55°C)	≥ 3 days	Days must be continuous



Turning

Source: @RedHookCompost



Turning



Hidden Harvest Farm, Baltimore. Source: ILSR



Howard University Community Compost Cooperative, Washington, DC. Source: Loop Closing



Turning

What does turning do?

- Charges the material with fresh air
- Releases trapped heat & moisture
- Distributes moisture, nutrients & organisms
- Breaks up and redistributes air channels
- “Fluffs” and shreds particles
- Breaks apart clumps of material
- Generally reinvigorates & provides a burst of microbial activity

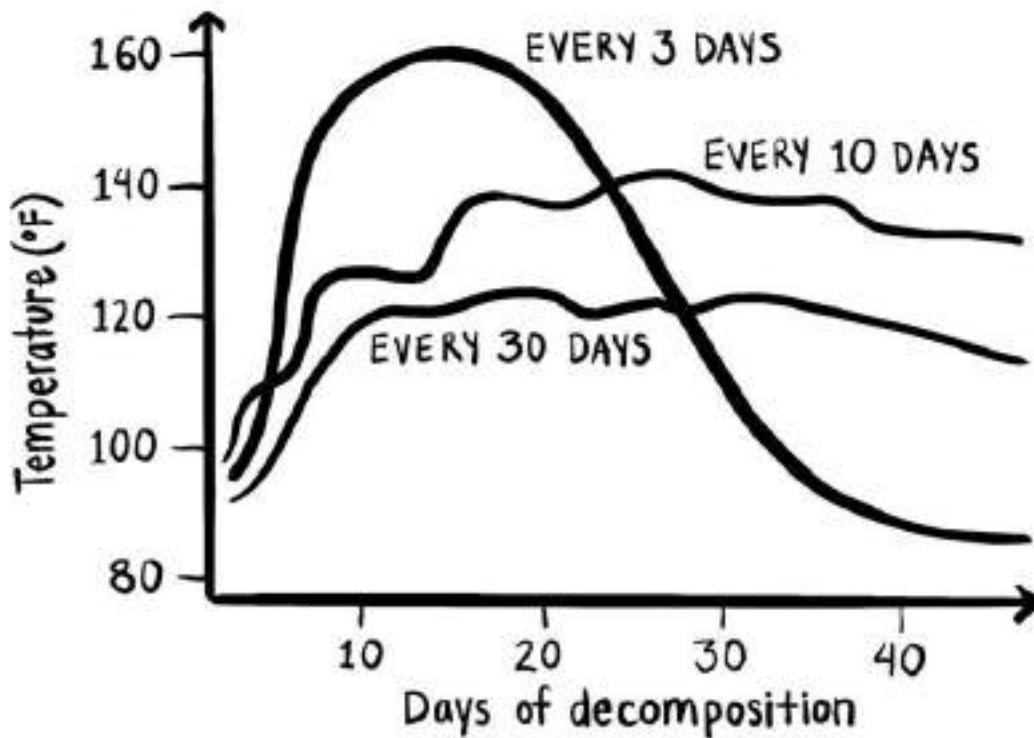


Rust Belt Riders's Nathan Rutz turns a thermophilic composting pile. Photo: Jacob Koestler.



Source: U.S. Composting Council, Foundation of Composting Workshop

TURNING FREQUENCY EFFECTS ON COMPOSTING



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Tools for Monitoring Your Compost

1. Your senses! (Nose, eyes, and hands)
2. Temperature sensor
3. Bucket & scale
4. Binder or online form



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Compost
Monitoring
Log by
Compost
Technical
Services:
<https://www.composttechnicalservices.com/resources>

Log originally developed by Highfields Center For Composting



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Keeping Records

BK ROT COMPOST LOG		PILE TEMPERATURES		
DAY/DATE	WORKER	BIN #	TEMP	ACTIONS/OBSERVATIONS
5/22/16	Domingo	Washo	100	used to Seal new pile
11	11	4	81	Sifted
11	11	3	94	moved over
5/24/16	DENISE/ Domingo/Chloe	2	140	moved over
11	11	1	156	started new
11	11	1	135	Turned/Watered/Added
11	11	2	148	11
11	11	3	144	11
11	11	4	118	11
11	11	Washo	150	watered

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Keeping Records

Food scrap drop-off bucket & hanging scale at the Howard University Community Compost Cooperative operated by Loop Closing



Compost Project Tracking Binder at Prospect Heights Community Garden in Brooklyn, NYC.

ILSR

Keeping Records

NSR Sample Compost Monitoring Form

All questions with red asterisk must be completed (the remaining questions are optional).

* Required

Date of Data *

Date

mm/dd/yyyy

Name *

Your answer

Location (bin 1, 2, 3, 4) *

Choose

«All questions below refer to the location (bin, chamber, or pile) selected above.»



Troubleshooting

Ideal Composting Conditions

Variable	Ideal Range
C:N Ratio	25:1 - 30:1
Moisture Content	50 - 60%
Particle Size	$\frac{1}{8}$ - 2 inches
Bulk Density	800 - 1,000 pounds per cubic yard
Pile Size	> 1 cubic yard
Temperature	120 - 150 °F



Troubleshooting



The hand-squeeze test



Measuring bulk density



Troubleshooting Moisture Content

Symptom	Problem	Solution
Hand is dry, material is crumbly	Not enough water	Moisten and turn pile
Water drips down arm	Too much water	Add high carbon material and turn pile



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Troubleshooting Bulk Density

Symptom	Problem	Solution
Low bulk density	Too light: too much bulking agent or too dry	Determine if moisture needs adjusting. Consider adding more bioavailable brown or greens. Remeasure bulk density.
High bulk density	Too dense: too much nitrogen or water, or too little bulking agent	Determine if moisture needs adjusting. Consider adding more bioavailable brown or bulking agent. Remeasure bulk density.

**Ideal bulk density range:
800 - 1,000
lbs / cubic
yard**



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Troubleshooting Continued

Symptom	Problem	Solution
Compost has bad odor	Too much nitrogen or water, or not enough air	Add high carbon and/or bulking material and turn pile. Remeasure moisture & bulk density.



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Troubleshooting Continued

Symptom	Problem	Solution
Center of pile is dry	Not enough water	Moisten and turn pile. Remeasure moisture & bulk density.
Pile is damp, but warm only in middle	Pile is too small	Add more material and mix into new pile. Remeasure moisture & bulk density.
Pile is damp, but will not heat up	Lack of nitrogen	Mix in nitrogen source like grass clippings or coffee grounds. Remeasure moisture & bulk density.



Troubleshooting Continued



The hand-squeeze test



Measuring bulk density



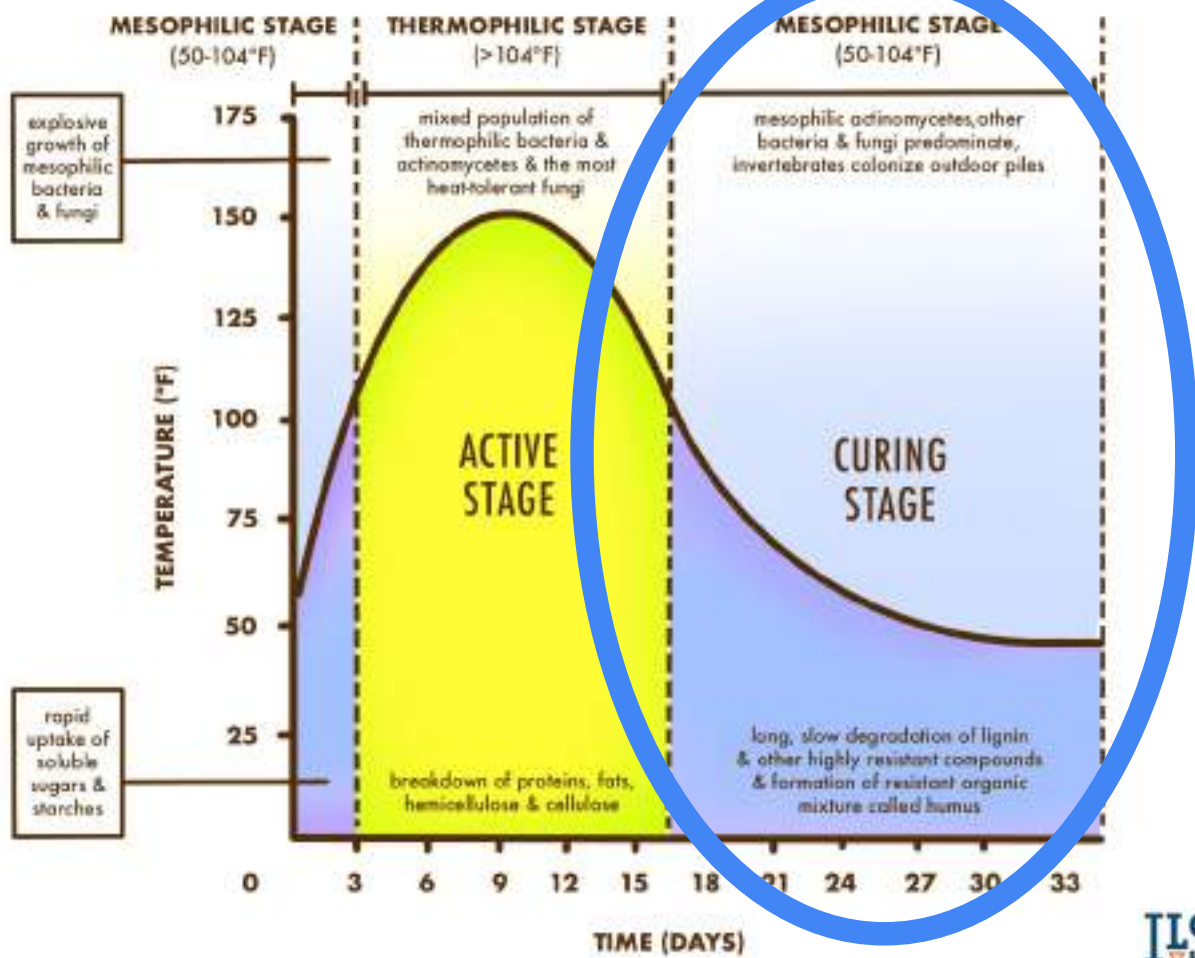
Finishing the Composting Process

- ✓ Curing
- ✓ Sifting or screening
- ✓ Compost testing
- ✓ Using finished compost



Volunteers sift compost at the Red Hook Community Farm Compost Operation in Brooklyn, NYC.
Source: @RedHookCompost





Curing



Curing demonstration at Earth Matter NY on Governors Island. Source: ILSR

Curing



Three bin composting system at the Takoma Compost Cooperative, part of the DC Department of Parks and Recreation's Community Compost Network. *Source: ILSR*



Curing



Loop Closing at the Howard University Compost Cooperative, part of the DC Department of Parks and Recreation's Community Compost Network.

Source: Loop Closing





**How do
you know
when it's
done?**

**How do you know
when your compost is finished?**

**Characteristics of Mature
Compost**



How do you know when your compost is finished?

Characteristics of Mature Compost

→ Color is dark brown



How do you know when your compost is finished?

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→ Color is dark brown

→ Crumbly and loose



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- Color is dark brown
- Crumbly and loose
- Has an earthy smell



How do you know when your compost is finished?

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- Color is dark brown
- Crumbly and loose
- Has an earthy smell
- Within 10°F of ambient temperature



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- Within 10°F of ambient temperature
- Pile is 1/3 of its original volume



How do you know when your compost is finished?

Characteristics of Mature Compost

- Color is dark brown
- Crumbly and loose
- Has an earthy smell
- Within 10°F of ambient temperature
- Pile is 1/3 of its original volume
- Contains no readily recognizable food scraps



Screening

Photo: ILSR



Photo: ILSR



Photo: @BK_ROT



Photo: ILSR



Screening

@RedHookCompost



Screening



Youth employees of the Baltimore Compost Collective sift compost.
Source: Marvin Hayes, Baltimore Compost Collective.



Sifter at ECUF Compost Project, East Capitol Urban Farm, DC.
Photo: ILSR



Trommel Screener



Custom built trommel screener at ECO City Farms in Bladensburg, MD. *Source: ILSR*



Trommel screener building at Malcolm House in Baltimore. *Source: ILSR*



Building a Trommel Screen

📅 POSTED JUNE 13, 2018 👤 BRIK MARTIN



<https://conscious-compost.com/building-a-small-trommel-screen/>



Compost Testing

✓ Why do it?

- You're a commercial facility that's required to
- You plan to sell your compost
- Your compost will be used in food production
- Your composting meat, manures and/or large amounts of food scraps

✓ Benefits

- Relationship with a reputable lab
- Assurance you have reduced pathogens
- Know how ready-to-use your compost is



Testing compost pH, electrical conductivity and seed germination during NSR Master Composter compost testing module.
Source: ILSR



Compost Testing

✓ Field Tests:

- “Ziplock” test
- Seed germination bioassay for phytotoxins
- Solvita for maturity
- Electrical conductivity for soluble salt content
- pH

✓ Lab Tests:

- Nutrient content
- Organic matter
- Trace metals
- Pathogens



Testing compost pH, electrical conductivity and seed germination during NSR Master Composter compost testing module.
Source: ILSR



Test Your Compost: Respiration Test

Respiration test is an indicator of compost stability

1. Add a handful of moist compost to a ziplock bag
2. Press out air and seal
3. Store out of the sun for 3 days
4. Open and smell



Ammonia = need to compost longer



Test Your Compost: Germination Test

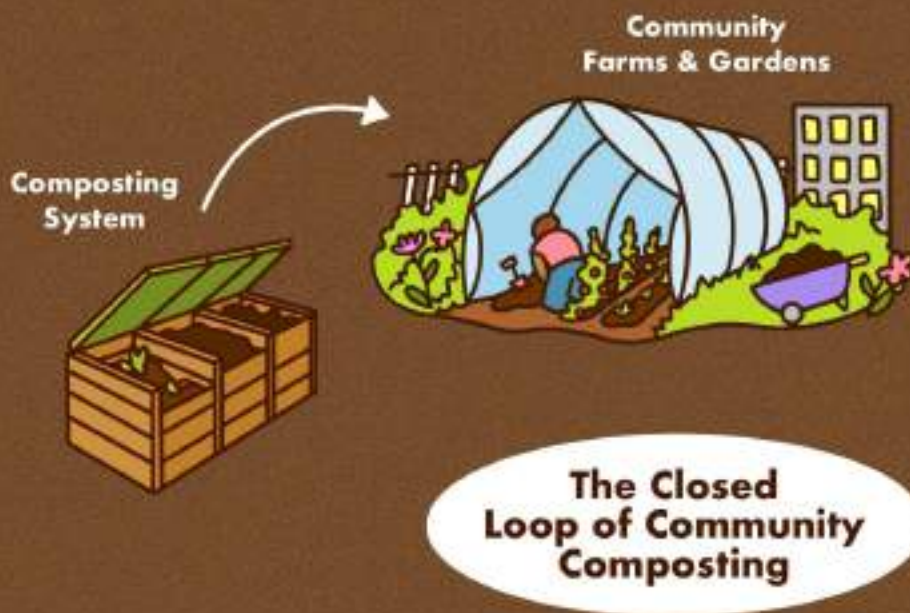
The germination test is an indicator of quality

1. Count 100 new seeds
2. Add to plate of moist compost
3. Keep compost moist and warm
4. Compare number of sprouts to germination rate on seed packet



Good germination = sign of good compost





Using Finished Compost

Soil Incorporation

Mix a 2-4 inch layer of compost into the top 6-8 inches of soil prior to seeding or transplanting.



Photo: ILSR



Sources: New York City Master Composter Course Manual (2015) & Cornell Waste Management Institute's Master Composter Resource Manual (1998)

Using Finished Compost

Soil Incorporation

Mix a 2-4 inch layer of compost into the top 6-8 inches of soil prior to seeding or transplanting.

Potting Mix (for seedlings in small containers)

- ✓ Sift compost through 1/4 inch mesh
- ✓ Mix two parts compost, one part coarse sand, and one part loamy soil or coconut coir
- ✓ Add 1/2 cup garden lime (pulverized limestone) per 8 gallons potting mix



Photo: ILSR



Photo:
@DirtCraft
Organics



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Mulch

Add mulch in a 3 inch layer on the soil surface.



Photo: ILSR



Photo:
@DirtCraft
Organics



Photo: The
Orchard
People



Sources: New York City Master Composter Course Manual (2015) & Cornell Waste Management Institute's Master Composter Resource Manual (1998)

COMMON STEPS IN THE COMMUNITY COMPOSTING PROCESS

- 1** Collect & store composting feedstocks



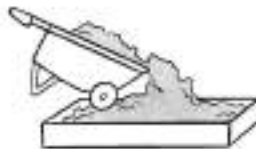
- 2** Record volume or weight of incoming materials



- 3** Chop up large feedstocks as needed



- 4** Mix feedstocks & build your compost pile



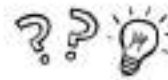
- 5** Monitor & record temperature, moisture level & odor



- 6** Mix & water regularly



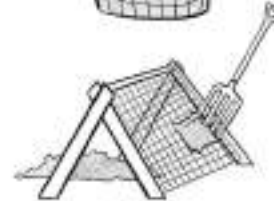
- 7** Troubleshoot as needed



- 8** Cure



- 9** Sift



- 10** Use or store finished compost



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Managing the Composting Process

- ✓ Managing the Active Composting Process
- ✓ Compost Troubleshooting
- ✓ Finishing the Composting Process



Source: @LA Compost

ILSR

Site Design & Management

- ✓ Navigating State & Local Permitting & Regulations
- ✓ Choosing a System
- ✓ Site Layout & Management Plan
- ✓ Health & Safety
- ✓ Tips to Avoid Rodents
- ✓ What Do You Notice?